Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
solve equations using the Subtraction and Addition Properties of Equality.			
solve equations that need to be simplified.			
translate an equation and solve.			
translate and solve applications.			

ⓑ *If most of your checks were:*

...confidently. Congratulations! You have achieved the objectives in this section. Reflect on the study skills you used so that you can continue to use them. What did you do to become confident of your ability to do these things? Be specific.

...with some help. This must be addressed quickly because topics you do not master become potholes in your road to success. In math, every topic builds upon previous work. It is important to make sure you have a strong foundation before you move on. Whom can you ask for help? Your fellow classmates and instructor are good resources. Is there a place on campus where math tutors are available? Can your study skills be improved?

...no—I don't get it! This is a warning sign and you must not ignore it. You should get help right away or you will quickly be overwhelmed. See your instructor as soon as you can to discuss your situation. Together you can come up with a plan to get you the help you need.

8.2 Solve Equations Using the Division and Multiplication Properties of Equality

Learning Objectives

By the end of this section, you will be able to:

- Solve equations using the Division and Multiplication Properties of Equality
- Solve equations that need to be simplified

☑ **BE PREPARED** 8.4 Before you get started, take this readiness quiz.

Simplify: $-7\left(\frac{1}{-7}\right)$.

If you missed this problem, review Example 4.28.

☑ **BE PREPARED** 8.5 What is the reciprocal of $-\frac{3}{8}$?

If you missed this problem, review Example 4.29.

☑ **BE PREPARED** 8.6 Evaluate $9x + 2$ when $x = -3$.

If you missed this problem, review Example 3.56.

Solve Equations Using the Division and Multiplication Properties of Equality

We introduced the Multiplication and Division Properties of Equality in Solve Equations Using Integers; The Division Property of Equality and Solve Equations with Fractions. We modeled how these properties worked using envelopes and counters and then applied them to solving equations (See Solve Equations Using Integers; The Division Property of Equality). We restate them again here as we prepare to use these properties again.

Division and Multiplication Properties of Equality

Division Property of Equality: For all real numbers a, b, c, and $c \neq 0$, if $a = b$, then $\frac{a}{c} = \frac{b}{c}$.

Multiplication Property of Equality: For all real numbers a, b, c, if $a = b$, then $ac = bc$.

When you divide or multiply both sides of an equation by the same quantity, you still have equality.

Let's review how these properties of equality can be applied in order to solve equations. Remember, the goal is to 'undo' the operation on the variable. In the example below the variable is multiplied by 4, so we will divide both sides by 4 to 'undo' the multiplication.

EXAMPLE 8.13

Solve: $4x = -28$.

Solution

We use the Division Property of Equality to divide both sides by 4.

	$4x = -28$
Divide both sides by 4 to undo the multiplication.	$\frac{4x}{4} = \frac{-28}{4}$
Simplify.	$x = -7$
Check your answer. Let $x = -7$.	
$4x = -28$	
$4(-7) \stackrel{?}{=} -28$	
$-28 = -28$ ✓	

Since this is a true statement, $x = -7$ is a solution to $4x = -28$.

TRY IT 8.25 Solve: $3y = -48$.

TRY IT 8.26 Solve: $4z = -52$.

In the previous example, to 'undo' multiplication, we divided. How do you think we 'undo' division?

EXAMPLE 8.14

Solve: $\frac{a}{-7} = -42$.

Solution

Here a is divided by -7. We can multiply both sides by -7 to isolate a.

	$\frac{a}{-7} = -42$
Multiply both sides by -7.	$-7\left(\frac{a}{-7}\right) = -7(-42)$ $\frac{-7a}{-7} = 294$
Simplify.	$a = 294$

Check your answer. Let $a = 294$.

$\frac{a}{-7} = -42$	
$\frac{294}{-7} \stackrel{?}{=} -42$	
$-42 = -42$ ✓	

TRY IT 8.27 Solve: $\frac{b}{-6} = -24$.

TRY IT 8.28 Solve: $\frac{c}{-8} = -16$.

EXAMPLE 8.15

Solve: $-r = 2$.

Solution

Remember $-r$ is equivalent to $-1r$.

	$-r = 2$
Rewrite $-r$ as $-1r$.	$-1r = 2$
Divide both sides by -1.	$\frac{-1r}{-1} = \frac{2}{-1}$
	$r = -2$

Check.	$-r = 2$	
Substitute $r = -2$	$-(-2) \stackrel{?}{=} 2$	
Simplify.	$2 = 2$ ✓	

In Solve Equations with Fractions, we saw that there are two other ways to solve $-r = 2$.

We could multiply both sides by -1.

We could take the opposite of both sides.

TRY IT 8.29 Solve: $-k = 8$.

TRY IT 8.30 Solve: $-g = 3$.

EXAMPLE 8.16

Solve: $\frac{2}{3}x = 18$.

Solution

Since the product of a number and its reciprocal is 1, our strategy will be to isolate x by multiplying by the reciprocal of $\frac{2}{3}$.

	$\frac{2}{3}x = 18$
Multiply by the reciprocal of $\frac{2}{3}$.	$\frac{3}{2} \cdot \frac{2}{3}x = \frac{3}{2} \cdot 18$
Reciprocals multiply to one.	$1x = \frac{3}{2} \cdot \frac{18}{1}$
Multiply.	$x = 27$
Check your answer. Let $x = 27$	
$\frac{2}{3}x = 18$	
$\frac{2}{3} \cdot 27 \stackrel{?}{=} 18$	
$18 = 18$ ✓	

Notice that we could have divided both sides of the equation $\frac{2}{3}x = 18$ by $\frac{2}{3}$ to isolate x. While this would work, multiplying by the reciprocal requires fewer steps.

TRY IT 8.31 Solve: $\frac{2}{5}n = 14$.

TRY IT 8.32 Solve: $\frac{5}{6}y = 15$.

Solve Equations That Need to be Simplified

Many equations start out more complicated than the ones we've just solved. First, we need to simplify both sides of the equation as much as possible

EXAMPLE 8.17

Solve: $8x + 9x - 5x = -3 + 15$.

Solution

Start by combining like terms to simplify each side.

	$8x + 9x - 5x = -3 + 15$
Combine like terms.	$12x = 12$

Divide both sides by 12 to isolate x.	$\frac{12x}{12} = \frac{12}{12}$
Simplify.	$x = 1$
Check your answer. Let $x = 1$	
$8x + 9x - 5x = -3 + 15$	
$8 \cdot 1 + 9 \cdot 1 - 5 \cdot 1 \stackrel{?}{=} -3 + 15$	
$8 + 9 - 5 \stackrel{?}{=} -3 + 15$	
$12 = 12$ ✓	

TRY IT 8.33 Solve: $7x + 6x - 4x = -8 + 26$.

TRY IT 8.34 Solve: $11n - 3n - 6n = 7 - 17$.

EXAMPLE 8.18

Solve: $11 - 20 = 17y - 8y - 6y$.

Solution

Simplify each side by combining like terms.

	$11 - 20 = 17y - 8y - 6y$
Simplify each side.	$-9 = 3y$
Divide both sides by 3 to isolate y.	$\frac{-9}{3} = \frac{3y}{3}$
Simplify.	$-3 = y$
Check your answer. Let $y = -3$	
$11 - 20 = 17y - 8y - 6y$	
$11 - 20 \stackrel{?}{=} 17(-3) - 8(-3) - 6(-3)$	
$11 - 20 \stackrel{?}{=} -51 + 24 + 18$	
$-9 = -9$ ✓	

Notice that the variable ended up on the right side of the equal sign when we solved the equation. You may prefer to take one more step to write the solution with the variable on the left side of the equal sign.

TRY IT 8.35 Solve: $18 - 27 = 15c - 9c - 3c$.

TRY IT 8.36 Solve: $18 - 22 = 12x - x - 4x$.

EXAMPLE 8.19

Solve: $-3(n - 2) - 6 = 21$.

Solution

Remember—always simplify each side first.

	$-3(n-2)-6=21$
Distribute.	$-3n+6-6=21$
Simplify.	$-3n=21$
Divide both sides by -3 to isolate n.	$\frac{-3n}{-3}=\frac{21}{-3}$ $n=-7$
Check your answer. Let $n=-7$.	
$-3(n-2)-6=21$	
$-3(-7-2)-6\stackrel{?}{=}21$	
$-3(-9)-6\stackrel{?}{=}21$	
$27-6\stackrel{?}{=}21$	
$21=21$ ✓	

TRY IT 8.37 Solve: $-4(n - 2) - 8 = 24$.

TRY IT 8.38 Solve: $-6(n - 2) - 12 = 30$.

LINKS TO LITERACY

The *Links to Literacy* activity, "Everybody Wins" will provide you with another view of the topics covered in this section.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Solving One Step Equation by Mult/Div. Integers (Var on Left) (https://www.openstax.org/l/24OneStepMultiL)
Solving One Step Equation by Mult/Div. Integers (Var on Right) (https://www.openstax.org/l/24OneStepMultiR)
Solving One Step Equation in the Form: $-x = -a$ (https://www.openstax.org/l/24xa)

SECTION 8.2 EXERCISES

Practice Makes Perfect

Solve Equations Using the Division and Multiplication Properties of Equality

In the following exercises, solve each equation for the variable using the Division Property of Equality and check the solution.

68. $8x = 32$

69. $7p = 63$

70. $-5c = 55$

71. $-9x = -27$

72. $-90 = 6y$

73. $-72 = 12y$

74. $-16p = -64$

75. $-8m = -56$

76. $0.25z = 3.25$

77. $0.75a = 11.25$

78. $-3x = 0$

79. $4x = 0$

In the following exercises, solve each equation for the variable using the Multiplication Property of Equality and check the solution.

80. $\frac{x}{4} = 15$

81. $\frac{z}{2} = 14$

82. $-20 = \frac{q}{-5}$

83. $\frac{c}{-3} = -12$

84. $\frac{y}{9} = -6$

85. $\frac{q}{6} = -8$

86. $\frac{m}{-12} = 5$

87. $-4 = \frac{p}{-20}$

88. $\frac{2}{3}y = 18$

89. $\frac{3}{5}r = 15$

90. $-\frac{5}{8}w = 40$

91. $24 = -\frac{3}{4}x$

92. $-\frac{2}{5} = \frac{1}{10}a$

93. $-\frac{1}{3}q = -\frac{5}{6}$

Solve Equations That Need to be Simplified

In the following exercises, solve the equation.

94. $8a + 3a - 6a = -17 + 27$

95. $6y - 3y + 12y = -43 + 28$

96. $-9x - 9x + 2x = 50 - 2$

97. $-5m + 7m - 8m = -6 + 36$

98. $100 - 16 = 4p - 10p - p$

99. $-18 - 7 = 5t - 9t - 6t$

100. $\frac{7}{8}n - \frac{3}{4}n = 9 + 2$

101. $\frac{5}{12}q + \frac{1}{2}q = 25 - 3$

102. $0.25d + 0.10d = 6 - 0.75$

103. $0.05p - 0.01p = 2 + 0.24$

Everyday Math

104. **Balloons** Ramona bought 18 balloons for a party. She wants to make 3 equal bunches. Find the number of balloons in each bunch, b, by solving the equation $3b = 18$.

105. **Teaching** Connie's kindergarten class has 24 children. She wants them to get into 4 equal groups. Find the number of children in each group, g, by solving the equation $4g = 24$.

106. **Ticket price** Daria paid $\$36.25$ for 5 children's tickets at the ice skating rink. Find the price of each ticket, p, by solving the equation $5p = 36.25$.

107. **Unit price** Nishant paid $\$12.96$ for a pack of 12 juice bottles. Find the price of each bottle, b, by solving the equation $12b = 12.96$.

108. **Fuel economy** Tania's SUV gets half as many miles per gallon (mpg) as her husband's hybrid car. The SUV gets 18 mpg. Find the miles per gallons, m, of the hybrid car, by solving the equation $\frac{1}{2}m = 18$.

109. **Fabric** The drill team used 14 yards of fabric to make flags for one-third of the members. Find how much fabric, f, they would need to make flags for the whole team by solving the equation $\frac{1}{3}f = 14$.